
VECTOR SPACE PROBLEM SHEET

1

- (a) Show that if the columns, $\mathbf{m}_j$, of an $n \times n$ matrix, $\mathbf{M}$, are not independent then $\mathbf{M}$ does not have a unique inverse. For the columns to not be independent implies there exists some set of coefficients c_j such that

$$\sum_{j=1}^n c_j \mathbf{m}_j = \mathbf{0}.$$

We note that the i^{th} element of $\mathbf{m}_j = M_{ij}$ so

$$\sum_{j=1}^n c_j M_{ij} = 0$$

We consider the vector $\mathbf{c}$ with elements c_i , then the i^{th} component of $\mathbf{M}\mathbf{c}$ is equal to

$$[\mathbf{M}\mathbf{c}]_i = \sum_{j=1}^n M_{ij} c_j = 0$$

or $\mathbf{M}\mathbf{c} = \mathbf{0}$. Consequently for any vector $\mathbf{x}$ and any scalar b we have that

$$\mathbf{M}(\mathbf{x} + b\mathbf{c}) = \mathbf{M}\mathbf{x}$$

if we denote $\mathbf{y} = \mathbf{M}\mathbf{x}$, then there is not a unique vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{M}\mathbf{x}$, but rather a family of vectors $\mathbf{x} + b\mathbf{c}$ for any b .

[3 marks]

- (b) Show that for three $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ the inverse of $\mathbf{ABC}$ is $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$. You can do this by multiplying the product of matrices, but its inverse and use associativity of matrix multiplication

$$\begin{aligned} (\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{ABC}) &= \mathbf{C}^{-1}\mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{BC} \\ &= \mathbf{C}^{-1}(\mathbf{B}^{-1}\mathbf{B})\mathbf{C} = \mathbf{C}^{-1}\mathbf{C} = \mathbf{I} \end{aligned}$$

[3 marks]

- (c) For non-square matrices, $\mathbf{A}$ and $\mathbf{B}$ we can have $\mathbf{AB} = \mathbf{I}$ while $\mathbf{BA} \neq \mathbf{I}$. Show this is the case for the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix} = \mathbf{I}_1$$

$$\mathbf{BA} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

[3 marks]

- (d) For finite-dimensional square matrices if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{BA} = \mathbf{I}$. Thus $\mathbf{B} = \mathbf{A}^{-1}$. This is not entirely trivial to prove. We break the proof down. Show that if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{B}$ is injective (one-to-one). This means that for two different vectors x and y then $\mathbf{B}x \neq \mathbf{B}y$. Equivalently $\mathbf{B}(x - y) \neq 0$, unless $x = y$, but as x and y are arbitrary the only vector for which $\mathbf{B}x = 0$ is the vector $x = 0$. Prove that if $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}x = 0$ then $x = 0$.

Suppose $\mathbf{B}x = 0$ then multiply by $\mathbf{A}$

$$\mathbf{AB}x = 0$$

$$\mathbf{I}x = 0$$

Thus $x = 0$.

[2 marks]

- (e) Next we want to show that in a finite dimensional space a square matrix that is injective is also surjective (onto). That is, if $\mathbf{B}$ is injective then for any vector $\mathbf{y}$ there exists a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$. The reason a finite dimensional square matrix that is injective is also surjective is not so obvious. To prove this we consider a set of n **basis** vectors that span the entire space $\mathbb{R}^n$. We call these $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Because these are independent the only set of numbers c_i such that $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ are when $c_1 = c_2 = \dots = c_n = 0$. Show that the set of vectors $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ form a set of basis vectors (i.e. they are linear independent). This is easily proved by contradiction.

Since $\mathbf{B}$ is injective and $\mathbf{v}_i \neq \mathbf{0}$ (since basis vectors are non-zero), clearly $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ are non-zero. The non-trivial part is to show these vectors are linear independent. Suppose this is not the case and there exists a set of coefficients c'_i such that

$$\begin{aligned}\sum_{i=1}^n c'_i \mathbf{v}'_i &= \mathbf{0} \\ \sum_{i=1}^n c'_i \mathbf{B}\mathbf{v}_i &= \mathbf{0} \\ \mathbf{B} \sum_{i=1}^n c'_i \mathbf{v}_i &= \mathbf{0}.\end{aligned}$$

But, as $\mathbf{B}$ is injective this implies

$$\sum_{i=1}^n c'_i \mathbf{v}_i = \mathbf{0}.$$

Since the vectors $\mathbf{v}_i$ are linear independent, so this is a contradiction.

[5 marks]

- (f) Use the fact that the vectors $\mathbf{v}'_i$ are linearly independent to show that, for any vector $\mathbf{y}$, we can find a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$

For any vector $\mathbf{y}$ can be represented by a weighted sum of n independent vectors. In particular we can write

$$\begin{aligned}\mathbf{y} &= \sum_{i=1}^n c_i \mathbf{v}'_i = \sum_{i=1}^n c_i \mathbf{B}\mathbf{v}_i \\ &= \mathbf{B} \left(\sum_{i=1}^n c_i \mathbf{v}_i \right) = \mathbf{B}\mathbf{x}\end{aligned}$$

where $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$.

[2 marks]

TURN OVER
Page 3 of 6

- (g) Using the fact that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}$ is surjective show $\mathbf{BA} = \mathbf{I}$

We note that for all x

$$\begin{aligned}\mathbf{AB}x &= x \\ \mathbf{BAB}x &= \mathbf{B}x\end{aligned}$$

But as $\mathbf{B}$ is surjective for any vector y there exists a vector x such that $y = \mathbf{B}x$. Thus,

$$\mathbf{BA}y = y,$$

but as this is true for any vector y it follows that $\mathbf{BA} = \mathbf{I}$.

[2 marks]

End of question 1

2 We want to compute the derivative of $\log(\det(\mathbf{X}))$ with respect to the matrix $\mathbf{X}$. Our strategy will be to use the fact that

$$\lim_{\epsilon \rightarrow 0} \frac{\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X}))}{\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}.$$

We therefore expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ to first order in ϵ , subtract $\log(\det(\mathbf{X}))$, divide by ϵ , rearrange and read off $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. However, rearranging in this case is quite complicated.

- (a) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ and $\log(ab) = \log(a) + \log(b)$ expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ as a sum of $\log(\det(\mathbf{X}))$ and another logarithm.
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$$\begin{aligned}\log(\det(\mathbf{X} + \epsilon \mathbf{U})) &= \log(\det(\mathbf{X}(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}))) \\ &= \log(\det(\mathbf{X}) \det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})) \\ &= \log(\det(\mathbf{X})) + \log(\det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}))\end{aligned}$$

[5 marks]

- (b) The hard part is to expand $\det(\mathbf{I} + \epsilon \mathbf{M})$. Write out $\det(\mathbf{I} + \epsilon \mathbf{M})$ explicitly for a 2 dimensional matrix and expand to first order in ϵ
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$$\begin{aligned}\det(\mathbf{I} + \epsilon \mathbf{M}) &= \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{pmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\ &= 1 + \epsilon(M_{11} + M_{22}) + O(\epsilon^2)\end{aligned}$$

[2 marks]

(c) For an $n \times n$ matrix $\mathbf{M}$ compute the determinant of

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} & \dots & \epsilon M_{1n} \\ \epsilon M_{21} & 1 + \epsilon M_{22} & \dots & \epsilon M_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon M_{n,1} & \epsilon M_{n,2} & \dots & 1 + \epsilon M_{nn} \end{pmatrix}$$

up to first order in ϵ . Recall we can compute the determinant a matrix $\mathbf{X}$ as

$$\det(\mathbf{X}) = \sum_{i=1}^n (-1)^j X_{1j} \det(\hat{\mathbf{X}}_{1j})$$

where $\hat{\mathbf{X}}_{1j}$ is the matrix obtained by removing row 1 and column j from the matrix $\mathbf{X}$.

Expanding the determinant we get a term equal to the product of the diagonal elements

$$\prod_{i=1}^n (1 + \epsilon M_{ii}) = 1 + \epsilon \sum_{i=1}^n M_{ii} + O(\epsilon^2)$$

all other terms in the determinant involve at least terms of order ϵ^2 so do not contribute to order ϵ . In consequence

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)$$

[5 marks]

(d) For a more elegant proof we can use the fact that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i \qquad \text{tr} \mathbf{M} = \sum_{i=1}^n \lambda_i$$

where λ_i are the eigenvalues of the matrix $\mathbf{M}$. We will prove these results later when we look at eigen-systems. Note that if $\mathbf{v}_i$ is an eigenvector of the matrix $\mathbf{M}$, with eigenvalue λ_i then it is an eigenvector of $\mathbf{I} + c\mathbf{M}$ with eigenvalue $1 + c\lambda_i$ since

$$(\mathbf{I} + c\mathbf{M})\mathbf{v}_i = \mathbf{v}_i + c\lambda_i \mathbf{v}_i = (1 + c\lambda_i) \mathbf{v}_i.$$

Using these results show that $\det(\mathbf{I} + \epsilon \mathbf{M}) = 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)$.

$$\begin{aligned} \det(\mathbf{I} + \epsilon \mathbf{M}) &= \prod_{i=1}^n (1 + \epsilon \lambda_i) = 1 + \epsilon \sum_{i=1}^n \lambda_i + O(\epsilon^2) \\ &= 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2) \end{aligned}$$

[3 marks]

TURN OVER
Page 5 of 6

(e) Using the results

$$\begin{aligned}\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X})) &= \log(\det(\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U})) \\ \det(\mathbf{I} + \epsilon \mathbf{M}) &= 1 + \epsilon \text{tr} \mathbf{M} + O(\epsilon^2)\end{aligned}$$

together with the fact that $\log(1 + x) = x + O(x^2)$, compute $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. You may use the fact that $\log(1 + x) = x + O(x^2)$ and $\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T$.

We note that

$$\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X})) = \log(1 + \epsilon \text{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon^2)) = \epsilon \text{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon^2) = \epsilon \text{tr} \mathbf{U}^T (\mathbf{X}^{-1})^T$$

Thus,

$$\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}} = (\mathbf{X}^{-1})^T$$

[5 marks]

End of question 2